

CHM 120

Titration for Acetic Acid in Vinegar

Final Report

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Lesson Titration for Acetic Acid in Vinegar
Institution Rock Valley College
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Course CHM 120
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Test Your Knowledge

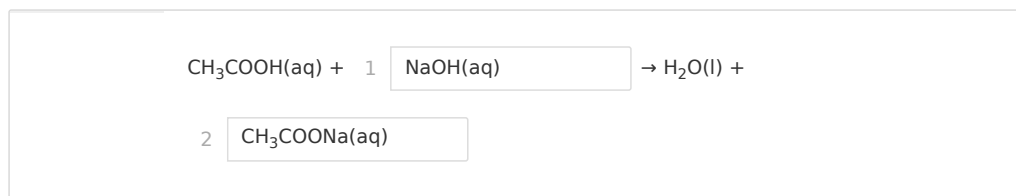
Match each term with the appropriate definition.

Completely ionizes in water	1	Strong acid
g/mol	2	Molar mass
(Grams of solute/grams of solution) x 100%	3	Mass percent
Moles of analyte = moles of titrant	4	Equivalence point
Known concentration in a titration	5	Titrant
Moles of solute/L of solution	6	Molarity
Only partially ionizes in water	7	Weak acid
Unknown concentration in a titration	8	Analyte
A quantitative, volumetric technique used to determine the concentration of an unknown analyte	9	Titration

Determine whether each chemical substance would remain the same color or turn pink in the presence of phenolphthalein.

No Color Change		Turns Pink	
1	Tap Water pH = 7	Lime Juice pH = 2	2
	Battery Acid pH = 1	Mashed Avocados pH = 6.5	Ammonia pH = 11
			Sea Water pH = 8.5

Complete the chemical equation.



Exploration

A weak acid completely ionizes in water to produce both a conjugate base and a H_3O^+ ion.

- ☐ True
- ☒ False

A titration is useful for determining the concentration of a given substance.

- ☐ True
- ☒ False

During a titration, a known concentration of ____ is added to a ____ of an unknown concentration.

- ☐ analyte; titrant
- ☒ titrant; analyte

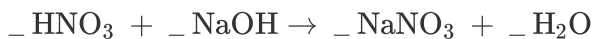
The equivalence point is the moment in a titration where exactly enough titrant has been added to completely react with the analyte.

- ☐ True
- ☒ False

Phenolphthalein is an effective pH indicator because equivalence points in titrations are marked by the analyte changing in color from ____ in acidic and neutral solutions, to ____ in basic solutions.

- ☐ light yellow; purple
- ☐ bright pink; colorless
- ☒ colorless; bright pink
- ☐ white; bright purple

What are the coefficients in the chemical equation?



- ☐ 1, 1, 2, 1
- ☐ 2, 2, 1, 3
- ☒ 1, 1, 1, 1
- ☐ 1, 3, 1, 1

Exercise 1

The manufacturer of the vinegar used in the experiment stated that the vinegar contained 5.0% acetic acid. What is the percent error between your result and the manufacturer statement? Show your work.

B / <u>U</u>	19 Word(s)
$\{[(\text{True Value} - \text{Observed Value})] / (\text{True Value})\} * 100\%$ True Value = 5.0% Observed Value = 3.0% $\{[(5.0 - 3.0)] / (5.0)\} * 100\% = 40\% \text{ Error}$	

What challenges would you encounter with the titration if you had used apple cider vinegar or balsamic vinegar as the analyte instead of white vinegar?

B / <u>U</u>	42 Word(s)
This could be due to white vinegar being more acidic than apple cider vinegar and/or balsamic vinegar. There will probably be not as much action taking place in the beaker due to the weak acid not having as much of an effect,	

If the tip of the syringe, "The Titrator", was not filled with NaOH before the initial volume reading was recorded, would the concentration of acetic acid in vinegar of that trial be greater than or less than the actual concentration? Please explain your answer.

B / <u>U</u>	45 Word(s)
It would probably be less than. I say this because the air bubbles that were in the tip would push out first with maybe just a bit of the NaOH, to where cutting the tip makes sure that it is straight NaOH coming out immediately.	

How would your experimentally determined vinegar concentration have differed if you had over-titrated (added drops of NaOH to the analyte beyond the stoichiometric equivalence point)?

B / U

35 Word(s)

Because more of the NaOH is being used, the volume used would change, which would change the result of the concentration of CH_3COOH in vinegar, and ultimately changing the percentage of CH_3COOH in the vinegar.

If a 7.0 mL sample of vinegar was titrated to the stoichiometric equivalence point with 7.5 mL of 1.5M NaOH, what is the mass percent of CH_3COOH in the vinegar sample? Show your work.

B / U

39 Word(s)

$(.0075)\text{L NaOH} * 1.5\text{mol NaOH} / \text{L NaOH} = .01125 \text{ mol NaOH}$

$.01125 / .0050 \leftarrow \text{the manufacturer's percent of acid in vinegar per L} = 2.25\text{mol/L}$

MASS PERCENT EQN: Plugging in values::

$2.25\text{mol CH}_3\text{COOH}/1000\text{mL} = .00225$

$.00225 * (7.0\text{mL}/1.006\text{g}) = .0156560636$

$(.0156560636 * 60.05\text{g CH}_3\text{COOH}) * 100\% = 94.\%$

Why is it important to do multiple trials of a titration, instead of only one trial?

B / U

47 Word(s)

To make sure that things are being observed and executed correctly for one. A second importance is that we can feel more safe about an assumption if we see something happen more than once. It is always important to do each experiment more than once for verification

Data Table 1: NaOH Titration Volume

	Initial NaOH Volume (mL)	Final NaOH Volume (mL)	Total Volume of NaOH Used (mL)
Trial 1	9mL	3mL	6mL
Trial 2	7mL	1mL	6mL
Trial 3	6mL	3.1mL	2.9mL
		Average Volume of NaOH Used (mL):	5.0mL

Data Table 2: Concentration of CH₃COOH in Vinegar

Average Volume of NaOH Used (mL)	Concentration CH ₃ COOH in vinegar (mol/L)	% CH ₃ COOH in vinegar
5.0mL	.5mol/L	3.0%

Competency Review

Titration is a quantitative, volumetric technique where a solution with a known concentration is added to a solution of an unknown concentration until the equivalence point is reached.

- ☒ True
- ☐ False

During a titration, the solution with the unknown concentration is called the ____.

- ☒ analyte
- ☐ titrant
- ☐ equivalence point

The moment in a titration where exactly enough titrant has been added to completely react with the analyte is called ____.

- ☐ the equivalence point
- ☐ the stoichiometric point
- ☒ both the equivalence point and the stoichiometric point
- ☐ none of the above

In the presence of a strong acid, phenolphthalein ____.

- ☐ remains colorless
- ☒ turns pink

When designing a titration between a titrant and an analyte, it is important to ensure the chemical equation is balanced.

- ☒ True
- ☐ False

A strong acid completely ionizes in water to produce ____.

- ☐ a conjugate acid and H^+ ions
- ☐ a conjugate base and H^+ ions
- ☒ a conjugate acid and OH^- ions
- ☐ a conjugate base and OH^- ions

During titration, a stopcock is used to ____.

- ☐ clamp and secure the titrator
- ☐ measure exactly 5 mL of titrant
- ☒ release one drop at a time from the titrator
- ☐ add NaOH to the titrator

The volume of titrant used during a titration will always be the exact same throughout three trials.

- ☐ True
- ☒ False

The concentration of acetic acid in vinegar is represented by the units ____.

- ☐ mL/mol
- ☒ mol/L
- ☐ mL
- ☐ mol

Extension Questions

A student did not read the directions to the experiment properly and mixed up where to place the NaOH solution and the vinegar. He put the vinegar in the titrator and the measured amount of NaOH in the beaker. He then added a drop of the phenolphthalein to the solution in the beaker.

Does the student need to empty out all of the solutions and start over again or can he go ahead and run the titration? If he runs the titration using the solutions as given above, what should he expect to see happen for results?

I believe the student will have to dump everything out and start over. I say this because the phenolphthalein should be reacting to become a solution with the vinegar and not the NaOH. This will most likely lead to different results. I wonder if the solution wouldn't just turn purple-pink because it is interacting with the NaOH so fast and so abundantly to where with the vinegar mixture, it has to fight its way through to interact and change color. This, I feel, explains why it changes color only for a brief second.